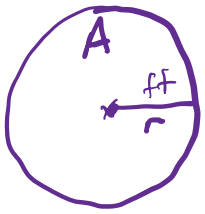
Problem 6. TIER 1 — THE FORMULA IS ONE YOU KNOW

A stone is dropped into a still pond, and the circular ripple's radius grows at 2 ft/s. How fast is the **area** enclosed by the ripple growing when the radius is 10 ft?

1. Draw + Label
2. What & when
3. Equation
4. Differentiate w.r.t. t
5. Sub + Solve

(1)



(2) $\frac{dr}{dt} = 2$

$\frac{dA}{dt} = ?$

when $r = 10$

(3) $A = \pi r^2$

(4) $\frac{d}{dt}A = \frac{d}{dt}\pi r^2$

$\frac{dA}{dt} = \pi 2r \frac{dr}{dt}$

(5) $\frac{dA}{dt} = 2\pi \cdot 10 \cdot 2 = 40\pi \frac{ft^2}{sec}$